End to End Machine Learning System For Wifi Congestion Prediction

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1. Problem Description

The Wifi Congestion Prediction System is a machine learning based application designed to predict network congestion in WiFi environments using traffic and performance related parameters. The system analyzes factors such as signal strength, bandwidth usage, number of connected devices, latency, and packet loss to identify congestion conditions and improve network performance.

The project uses important machine learning concepts including:

- Data Collection and Preprocessing
- Feature Engineering
- Classification Algorithms
- Model Evaluation and Prediction

2. Architecture

The system architecture follows these stages:

- Data Collection: Collecting WiFi traffic data from datasets or simulated network environments.

- Data Preprocessing: Handling missing values, normalization, encoding categorical values, and removing duplicates.

- Feature Selection: Selecting important parameters such as signal strength, latency, and bandwidth usage.

- Train-Test Split: Dividing data into training and testing sets.

- Model Training: Training machine learning models for congestion prediction.

- Prediction System: Predicting whether the network is congested or not.

- **Model Training:** Implementing classification models to learn network traffic patterns and detect congestion conditions.

- **Model Prediction:** Using the trained model to predict whether the WiFi network is congested or not based on real-time network parameters..

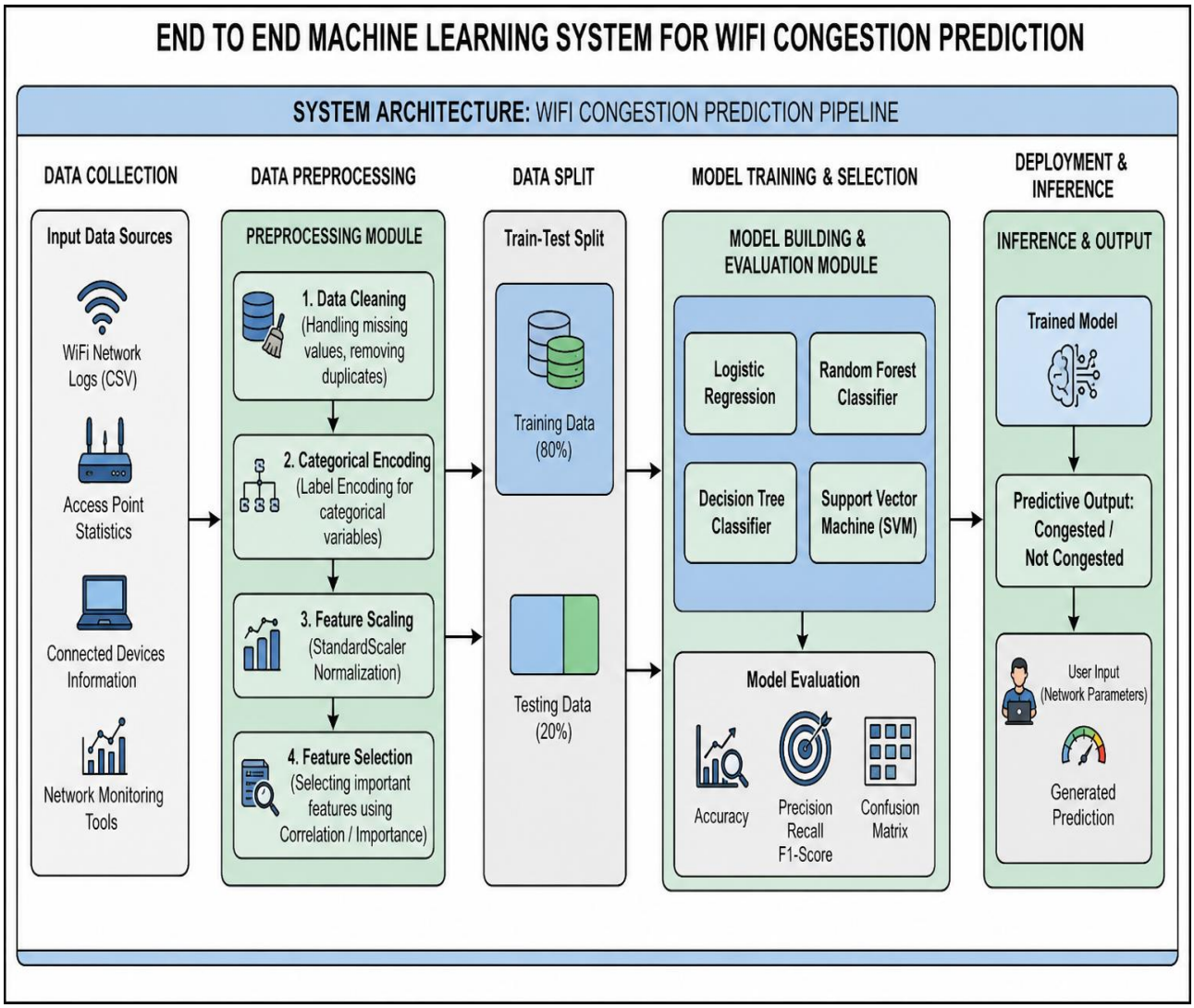


Fig.1 Sample flow diagram.

3. <u>Models Used(Applied/ Planned) Give a brief introduction ofthe models used</u>

The project implements supervised learning techniques specifically suited for continuous value prediction:

- **Logistic Regression:** Used for binary classification of congestion status..
- **Random Forest:** Improves prediction accuracy using multiple decision trees..

- **Support Vector Machine(SVM):** Separates congested and non-congested classes effectively.

4. Evaluation metrics used

To ensure high accuracy and reliability, the models are benchmarked using:

- **Accuracy Score:** Measures overall prediction accuracy.
- **F1 Score:** Balances precision and recall.
- **Confusion Matrix:** Shows correct and incorrect predictions.
- **Recall Score:** Measures ability to detect congestion cases.

5. Preliminary Results and Visualizations

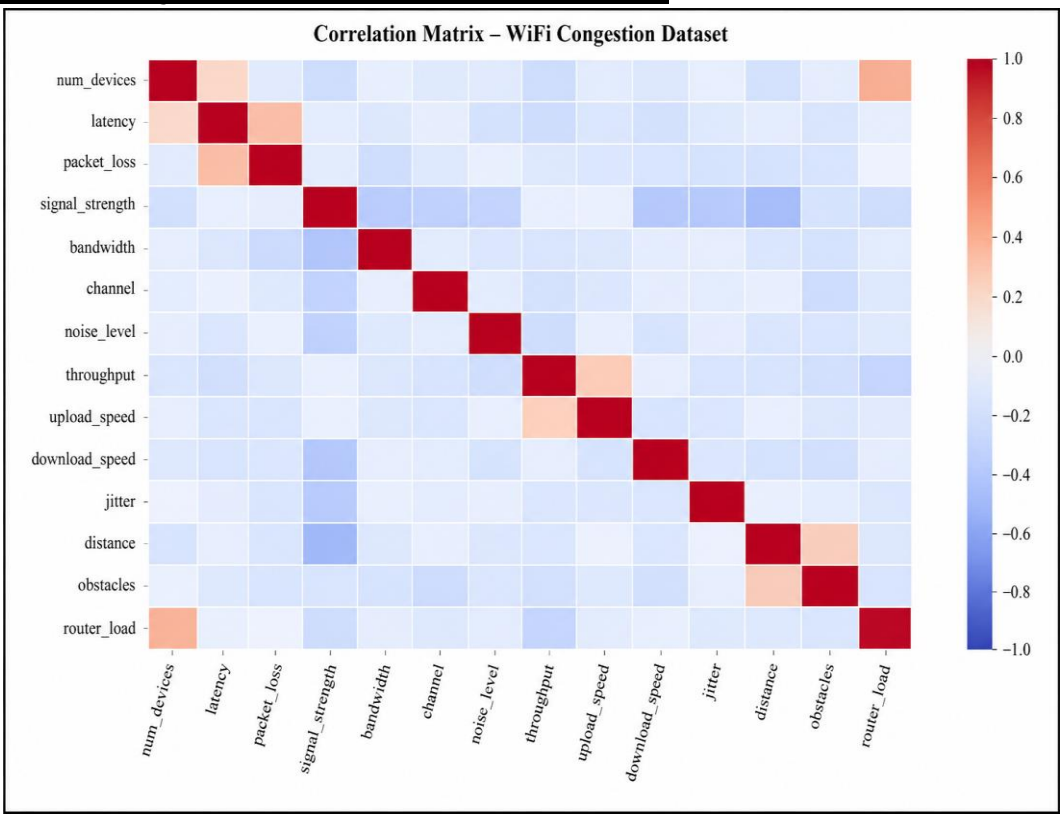
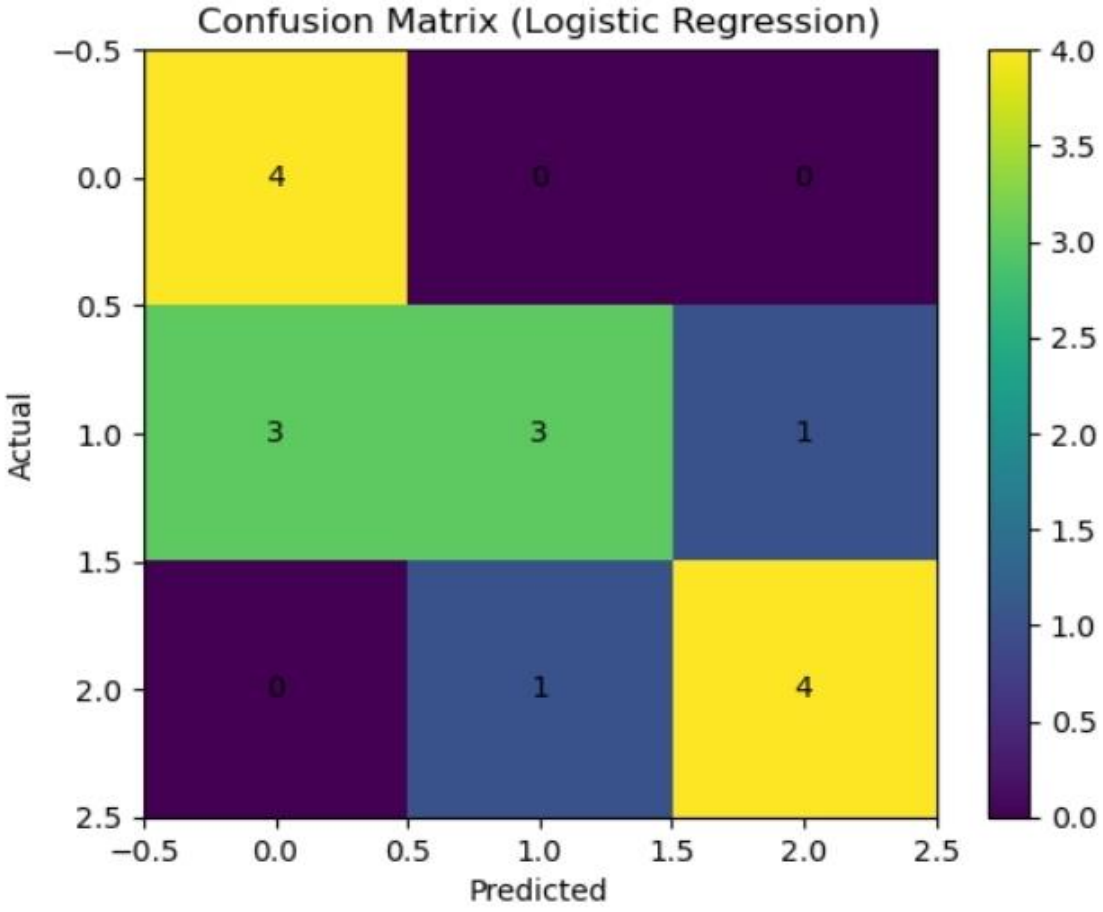


Fig.2 Confusion Matrix(Logictic Regression)



Classification Report

(WiFi Congestion Prediction)

	precision	recall	f1-score	support
0	0.57	1.00	0.73	4
1	0.75	0.43	0.55	7
2	0.80	0.80	0.80	5
accuracy			0.69	16
macro avg	0.71	0.74	0.69	16
weighted avg	0.72	0.69	0.67	16

Fig.4 Classification Report

- **Correlation Heatmap:** isualizes the relationship between WiFi network parameters such as latency, packet loss, signal strength, throughput, and router load. It helps identify features that strongly affect network congestion.

- **Confusion Matrix:** Displays the performance of the classification model by comparing actual congestion classes with predicted classes. It helps evaluate correct and incorrect predictions

- **Classification Report:** Provides important evaluation metrics such as precision, recall, F1-score, and accuracy to measure the effectiveness of the WiFi congestion prediction model.

6. References:

- **Books:**

- *Let Us Python*–YashavantKanetkar.
- Python Machine Learning – Sebastian Raschka

- **Online Resources:**

- Scikit-Learn Documentation
- GeeksforGeeks – Machine Learning Algorithms
- Kaggle Datasets for WiFi Congestion Analysis
- ProjectPro – Machine Learning Workflow Steps
- Python Official Documentation
- Research articles related to WiFi Congestion Prediction